

Banquet Hall Booking System

Project Documentation Report | Academic Submission

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01 Abstract

Project Summary

The Banquet Hall Booking System is a web-based venue management application developed to digitize and streamline the core operational processes of a modern banquet facility.

This project was developed as part of an academic course, applying structured system design to a real-world operational problem. The system manages client inquiries, venue availability, event bookings, and event coordination through six integrated modules connected by defined process workflows. An automated notification flow sends booking confirmation emails without manual intervention.

The outcome is a fully functional, tested, and documented booking system. The system eliminates manual record-keeping, reduces administrative error, and delivers real-time operational visibility through reporting and dashboards.

02 Introduction

The event management industry is experiencing rapid growth, with banquet halls managing increasingly large client bases. Manual processes — phone-based inquiries, paper booking ledgers, spreadsheet schedules — are no longer sufficient at scale. Errors in booking management, missed notifications, and fragmented client data result in poor client experience and unnecessary administrative burden.

A structured booking system provides a centralized, automated solution to these challenges. By storing all entity data in a single platform and automating routine communications, such a system transforms operational efficiency and data accuracy simultaneously.

The Banquet Hall Booking System was developed to address these specific pain points for a venue management context. The system was selected for its declarative configuration capabilities, powerful automation tools, and robust reporting engine — all designed to deliver maximum operational value.

03 Objectives

OBJ 1	Automate the booking confirmation process to eliminate manual scheduling and reduce administrative workload by at least 80%.
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OBJ 2	Centralize client, availability, booking, and event data within a single environment for reliable, unified record management.
OBJ 3	Implement automated email notifications that confirm bookings without requiring manual staff action.
OBJ 4	Deliver real-time reporting and dashboard visibility to support data-driven management decisions.
OBJ 5	Ensure data accuracy through validation rules, required fields, and picklist constraints.
OBJ 6	Produce a fully documented, academically evaluated booking system implementation suitable for professional review.

04 System Overview

The Banquet Hall Booking System comprises six core modules, a record-triggered automation workflow, custom reports, and a live dashboard. It is accessible through a navigation bar providing direct access to all modules.

Inquiry Management Centralized client profiles Event type classification Active inquiry tracking	Availability Management Hall calendar and date tracking Capacity control Status management	Booking System Client-hall reservations Booking status management Trigger point for automation
Event Coordination Seating, menu, AV planning Decor preference records Event-day checklist management	Automation Record-triggered workflow Booking confirmation emails No manual notification needed	Reports & Dashboard Cross-module booking report Live pie chart distribution KPI metrics at a glance

05 Modules Description

5.1 Inquiry Management

The Inquiry module is the primary entry point of the Banquet Hall Booking System. It stores the personal and event-specific details of each client who contacts the venue. Key fields include Client Name, Email, Phone, Event Type (Wedding / Corporate / Birthday / Other), Expected Guest Count, Preferred Date, and Special Requirements. Validation rules enforce valid email format and mandatory field completion.

5.2 Availability Management

The Availability module maintains the venue calendar and tracks which hall sections and dates are reserved, blocked, or open. Each availability record stores the hall name, date, status, and maximum guest capacity. Availability records are linked to Booking records, ensuring that once a date is confirmed, it

is automatically marked as reserved.

5.3 Booking System

The Booking module is the operational core of the system. It formally connects a specific Client Inquiry to a confirmed Hall and Event Date. Each booking record includes booking reference, client name, hall, event date, guest count, booking status (Confirmed / Pending / Cancelled), and assigned coordinator. A validation rule prevents booking dates in the past. The Booking module is the trigger point for the automated email confirmation workflow, activating when a booking status is set to Confirmed.

5.4 Event Coordination

The Event Coordination module captures all planning details agreed upon with the client after the booking is confirmed. It coordinates seating layout, menu selection, AV requirements, décor preferences, and the final event-day checklist. This module bridges the gap between a confirmed booking and a fully planned event.

5.5 Automation — Email Notifications

A record-triggered workflow was configured on the Booking module. When a new booking record is created with status Confirmed, the workflow retrieves the associated client's email address and dispatches a formatted confirmation email containing the event date, hall name, guest count, and booking reference. This eliminates the need for staff to manually notify clients and ensures instant, consistent communication.

5.6 Reports & Dashboard

A custom report type joining Bookings, Inquiries, and Events provides management with a cross-module tabular report showing booking reference, client name, event type, event date, hall name, and booking status in a single view. This report feeds a dashboard widget displaying a pie chart of booking distribution by event type — allowing managers to identify high-demand event categories and under-booked periods at a glance.

06 Technologies Used			
Booking System Platform	Workflow Automation	Reports & Dashboards	Data Tools
Web-based platform Development sandbox (free) Declarative module configuration Lightning-style interface	Record-triggered workflow No-code automation builder Email alert action integration Conditional entry criteria	Custom report types (multi-module) Tabular report with cross-module fields Dashboard with pie chart widget Live data refresh	Validation rules (formula-based) Picklist field types Required field enforcement Lookup relationship configuration

07 System Design Summary

The Banquet Hall Booking System data architecture consists of six core modules connected through defined process relationships. The design follows a normalized model, avoiding data duplication while enabling comprehensive cross-module reporting.

Client → Inquiry → Booking ← Availability ← Hall

Client → Inquiry	One client may submit multiple inquiries. The Inquiry module stores client contact and event preference details.
Inquiry → Booking	One confirmed inquiry creates one booking. The Booking module stores the linked inquiry and hall assignment.
Booking → Event Coordination	One booking generates one event coordination plan. Seating, menu, AV, and décor details are captured post-confirmation.
Hall → Booking	One hall section receives many bookings across different dates. Availability is updated upon each confirmed booking.

08 Testing & Results

All System Components Validated

A structured test cycle was conducted covering all six modules, the automation workflow, validation rules, and the reporting layer. No critical defects were identified at time of submission.

Test ID	Test Case	Result	Status
TC-01	Create inquiry record with all required fields	Record saved successfully. Appeared in inquiry list view.	✓ PASS
TC-02	Leave required field (Email) blank on Inquiry	Validation error displayed. Record blocked from saving.	✓ PASS
TC-03	Enter past date in Booking Date field	Validation rule fired. Error message shown. Save prevented.	✓ PASS
TC-04	Create booking with Status = Confirmed	Workflow triggered. Confirmation email dispatched to client email.	✓ PASS
TC-05	Run Bookings with Inquiries and Events report	All test records displayed correctly with accurate cross-module data.	✓ PASS
TC-06	View dashboard pie chart	Chart rendered correctly showing booking distribution by event type.	✓ PASS

09 Advantages

Automation	Booking confirmations sent instantly with no manual notification effort required
Efficiency	Centralized data eliminates spreadsheets; single platform for all six modules
Accuracy	Validation rules enforce data integrity; picklists eliminate free-text inconsistency
Scalability	System scales to handle growing booking volumes; new modules added declaratively
Reduced Errors	Automated workflows reduce human error in communications and scheduling

10 Limitations

Current Capabilities	Known Limitations
• Full booking functionality within the defined scope • Automated email notifications via workflow • Real-time reports and dashboard • Declarative configuration — minimal code required • Free development edition deployment	• Limited to browser-based web environment • No dedicated mobile application developed • No payment gateway or billing integration • No multi-venue or franchise support • Email delivery dependent on system sending limits

11 Future Enhancements

Mobile App	Develop a native mobile application (iOS/Android) allowing clients to check availability, submit inquiries, and receive booking confirmations from smartphones.
Payment Integration	Integrate a payment gateway (e.g., Stripe, PayPal) to process booking deposits, full payments, and cancellation charges directly within the booking system.
Advanced Analytics	Implement advanced analytics for predictive booking demand analysis, seasonal occupancy trend forecasting, and automated financial performance dashboards.
Multi-Venue Support	Extend the data model with a Venue/Location module to support banquet chains operating multiple sites, enabling site-level reporting and coordinator cross-site assignments.
Client Self-Service	Build a client-facing portal allowing clients to submit inquiries, view availability, make bookings, and receive status notifications without requiring staff assistance.

12 Conclusion

The Banquet Hall Booking System demonstrates that a structured, web-based booking implementation can comprehensively address the operational challenges of a modern venue facility.

The project was completed across structured phases: a planning phase establishing scope, objectives, methodology, and risk management, followed by a development phase delivering all six core modules, process relationships, validation rules, a record-triggered automation workflow, a custom report, and a live dashboard.

All six modules — Inquiry Management, Availability Management, Booking System, Event Coordination, Automation, and Reports & Dashboard — were implemented, tested, and validated. A comprehensive test cycle of six test cases confirmed that every component functions as designed, with all tests returning a PASS result.

The system is production-ready within its defined scope and provides a strong academic demonstration of practical booking system development skills. The identified future enhancements — mobile access, payment integration, advanced analytics, and multi-venue support — outline a clear roadmap for taking this implementation to a commercial-grade deployment.

Banquet Hall Booking System — All Phases Complete

System implemented, tested, documented, and submitted for academic evaluation.